

Environmental Sound Recognition Under Unseen and Adverse Acoustic Conditions

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Abstract

Environmental sound recognition (ESR) is essential for autonomous acoustic monitoring, yet out-of-distribution acoustic corruptions severely degrade real-world performance. In this work, we present a systematic multi-label benchmarking study on FSD50K under both clean and heavily corrupted acoustic conditions. We evaluate Convolutional Neural Network baselines (ResNet-18/34, EfficientNet-B0/B4) against Audio Spectrogram Transformers (ViT-Tiny/Small). While convolutional architectures like EfficientNet-B4 achieve the strongest clean precision (0.5396 mAP) and parameter efficiency, Vision Transformers provide competitive discrimination and Top-1 retrieval (76.54% for ViT-Small). Under unseen -5 dB SNR noise corruption, undefended ViT-Tiny performance degrades from 0.5068 to 0.2040 mAP. To address this degradation, we develop an acoustic augmentation pipeline combining continuous spectral-slope colored noise ($1/f^\alpha$, $\alpha \in [-2, 2]$), Schroeder room reverberation, Mixup ($\alpha = 0.5$), and learned attention pooling. The resulting robust model improves -5 dB performance to 0.3182 mAP (+56.0% relative gain over baseline ViT-Tiny) while reaching 0.5303 clean mAP without external noise datasets. Finally, extensive ablations quantify the trade-offs of token pooling heads, decision threshold calibration, and individual augmentation components.

Keywords: Environmental Sound Recognition, Audio Spectrogram Transformer, Noise Robustness, Digital Signal Processing Augmentation, FSD50K, Multi-Label Classification.

1 Introduction

Environmental Sound Classification (ESC) and multi-label Sound Event Recognition (SER) aim to identify and categorize sound events occurring in unstructured acoustic environments. Unlike human speech or Western musical compositions, environmental sounds are highly non-stationary, exhibit diverse spectral-temporal morphology (ranging from sharp acoustic transients to broad-band ambient hums), and frequently overlap in polyphonic mixtures.

1.1 The Real-World Acoustic Domain Gap

While deep learning models have achieved impressive recognition benchmarks on curated, clean audio datasets such as FSD50K [2] and AudioSet [3], real-world acoustic deployments present a severe domain shift. In applications such as Internet of Things (IoT) acoustic monitoring,

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hearing prosthetics, and autonomous navigation, incoming acoustic signals are continuously contaminated by ambient background noise (traffic, HVAC systems, wind, crowds). Most existing academic benchmarks evaluate models exclusively on identically distributed clean splits, obscuring the catastrophic failure modes that occur under adverse Signal-to-Noise Ratios (SNR).

1.2 Research Questions

To systematically bridge this gap, this investigation is structured around four concrete Research Questions (RQs):

- **RQ1 (Architectural Inductive Bias):** How do modern Vision Transformer (ViT) architectures compare against standard Convolutional Neural Networks (ResNet, EfficientNet) in multi-label environmental sound classification regarding recognition accuracy (mAP as primary metric, Top-1 hit and mAUC secondary), parameter efficiency, and computational cost?
- **RQ2 (Token Pooling Representations):** Given that Vision Transformers produce hundreds of spatial patch tokens over an audio spectrogram, how do different pooling mechanisms (Dual-Token CLS + DIST pooling vs. Global Average Pooling vs. Attention Pooling) affect multi-label sound discrimination?
- **RQ3 (Acoustic Noise Robustness):** How severely does model performance degrade under varying signal-to-noise ratios (SNR from +10 dB down to -5 dB), and can purely synthetic DSP augmentations (continuous colored noise, Schroeder reverberation) induce acoustic invariance without relying on external noise data?
- **RQ4 (Augmentation Component Contributions):** What are the individual and synergistic contributions of time-masking, frequency-masking, and Mixup in regularizing multi-label spectrogram classification, and how does decision threshold calibration resolve tail-class recall compression?

2 Related Work

2.1 CNNs vs. Transformers in Audio Classification

Historically, deep audio recognition relied on 2D Convolutional Neural Networks applied to time-frequency representations (Mel-spectrograms). Architectures such as PANNs (CNN-14) [6] and ResNet [5] treat spectrograms as single-channel images, leveraging local receptive fields and translation equivariance.

Recently, the Audio Spectrogram Transformer (AST) [1] adapted Vision Transformers (ViT) and Data-efficient Image Transformers (DeiT) [4] to audio. By dividing spectrograms into overlapping 2D patches and employing multi-head self-attention, AST captures non-local temporal-spectral dependencies across distant frequency bands (such as fundamental pitches and harmonic overtones).

2.2 Data Augmentation in Audio

Spectrogram data augmentation is essential for regularizing high-capacity neural networks. SpecAugment [7] applies random blocks of frequency masking and time masking, preventing models from

over-fitting to localized spectral artifacts. Mixup [8] performs convex combinations of audio pairs and their corresponding multi-hot ground truth labels:

$$\tilde{\mathbf{x}} = \lambda \mathbf{x}_i + (1 - \lambda) \mathbf{x}_j, \quad \tilde{\mathbf{y}} = \lambda \mathbf{y}_i + (1 - \lambda) \mathbf{y}_j \quad (1)$$

where $\lambda \sim \text{Beta}(\alpha, \alpha)$. In multi-label sound event recognition, Mixup naturally reflects the physical principle of acoustic wave superposition.

2.3 Robustness and Acoustic Corruption Benchmarks

Although speech recognition has standardized noise benchmarks (such as NOISEX-92 and the ACE Corpus [9]), open-domain environmental sound recognition lacks standardized out-of-distribution noise evaluation pipelines. Room reverberation modeling via statistical impulse responses [10] offers a principled, physics-informed mechanism to simulate reverberant propagation. Prior works often report accuracy exclusively on clean test splits, failing to quantify model robustness when tested under real acoustic degradation.

3 Dataset and Acoustic Environment Benchmark

3.1 Primary Dataset: FSD50K

We conduct all primary model training and validation on FSD50K [2] (Version 1.0), an open dataset of human-labeled sound events drawn from the AudioSet Ontology:

- **Official URL:** <https://zenodo.org/record/4060432>
- **Vocabulary:** 200 sound classes hierarchically organized into musical instruments, human sounds, domestic sounds, and environmental soundscapes.
- **Split Partitions:** Standard evaluation split consisting of a Development set (36,796 training clips, ~ 80 hours; 4,170 validation clips, ~ 12 hours) and an Evaluation test set (10,231 clips, ~ 24 hours).

3.2 Acoustic Noise Benchmark: ACE Dataset

To establish a rigorous out-of-distribution evaluation benchmark without contaminating training splits, we utilize the Acoustic Characterization of Environments (ACE) corpus [9] (<http://ee.ic.ac.uk/naylor/ACEweb/index.html>). The ACE corpus provides high-fidelity ambient background noises recorded across diverse indoor acoustic spaces (offices, lecture halls, meeting rooms) containing genuine noise sources such as HVAC ventilation fans, ambient human speech babble, and computer cooling equipment. We construct a standardized out-of-distribution evaluation suite by mixing the 10,231 clean held-out FSD50K evaluation clips with calibrated ACE ambient noises across four standardized Signal-to-Noise Ratio (SNR) tiers: +10 dB (mild interference), +5 dB (moderate interference), 0 dB (equal power), and -5 dB (heavy noise corruption), resulting in 40,924 corrupted evaluation instances.

3.3 Audio Preprocessing Pipeline

All audio signals are standardized to 16,000 Hz mono PCM audio. Clips are framed into fixed 10.0 s windows (160,000 samples), using random cropping during training and center cropping during validation/testing. Clips shorter than 10.0 s are zero-padded.

Log-Mel spectrograms are extracted using Short-Time Fourier Transform (STFT):

- Window length: 25 ms (400 samples), Hop length: 10 ms (160 samples).
- Mel filterbank: $N_{\text{mels}} = 128$ bins covering 0 Hz to 8,000 Hz.
- Logarithmic scaling: $S(t, f) = \log(\text{Mel}(t, f) + 10^{-6})$.
- Global standardization: $\tilde{S} = \frac{S - \mu}{\sigma}$ with global dataset statistics ($\mu = -4.2677$, $\sigma = 4.5690$). Target input tensor shape is $1 \times 128 \times 1000$.

4 Methodology

4.1 Convolutional Baseline Architectures vs. Audio Spectrogram Transformers

Audio spectrograms represent acoustic energy across continuous time (horizontal axis) and frequency (vertical axis). Treating a log-Mel spectrogram as an optical image introduces fundamental physical discrepancies that differentiate Convolutional Neural Networks from Vision Transformers.

In optical computer vision, 2D convolution leverages spatial translation equivariance: an object retains its identity regardless of spatial coordinates. In audio spectrograms, the temporal axis is translation-invariant (a sound event occurring at $t = 2$ s vs. $t = 6$ s retains its category), but the frequency axis is **strictly non-shift-invariant**. Shifting spectral energy upward alters pitch, formant structure, and harmonic intervals, fundamentally changing sound identity. Standard CNNs (ResNet-18/34, EfficientNet-B0/B4) apply small 3×3 convolutional kernels that excel at localized spectro-temporal textures and onset bursts, but require deep layer hierarchies and spatial pooling to capture multi-octave harmonic coherence (e.g., fundamental pitches at 100 Hz and integer overtones at 1–4 kHz).

In contrast, the Audio Spectrogram Transformer (AST) divides the 2D spectrogram into patches and applies multi-head self-attention. This establishes an $\mathcal{O}(1)$ information path length between any pair of time-frequency patches, allowing early self-attention layers to directly attend to simultaneous harmonic overtones across the entire 128 Mel channels. Learnable positional embeddings encode both temporal offset and frequency band index, capturing non-shift-invariant spectral structures naturally. To transfer visual pre-training without RGB channel expansion, the initial convolutional layer (`conv1`) of ResNet and EfficientNet is adapted from 3 channels to 1 single spectrogram channel by averaging pre-trained weights across the channel dimension:

$$\mathbf{W}_{\text{conv1_new}} = \frac{1}{3} \sum_{c=1}^3 \mathbf{W}_{\text{orig}}[:, c, :, :].$$

4.2 Main Proposed Method: Audio Spectrogram Transformer

Our primary classification architecture is the Audio Spectrogram Transformer (AST) built upon DeiT ViT-Tiny and ViT-Small backbones:

1. **Patch Tokenization:** The 2D spectrogram $\mathbf{X} \in \mathbb{R}^{128 \times 1000}$ is divided into $P \times P$ patches (16×16) with a stride of 10 pixels (an overlap of 6 pixels in both time and frequency), producing $N = 1,188$ patch tokens.
2. **Linear Projection and Positional Embeddings:** Each patch is flattened and linearly projected into dimension D ($D = 192$ for ViT-Tiny, $D = 384$ for ViT-Small):

$$\mathbf{z}_0 = [\mathbf{x}_{\text{CLS}}; \mathbf{x}_{\text{DIST}}; \mathbf{E}\mathbf{x}_p^1; \dots; \mathbf{E}\mathbf{x}_p^N] + \mathbf{E}_{\text{pos}} \quad (2)$$

where $\mathbf{x}_{\text{CLS}}$ and $\mathbf{x}_{\text{DIST}}$ are learnable classification and distillation tokens, and $\mathbf{E}_{\text{pos}} \in \mathbb{R}^{(N+2) \times D}$ represents 1D learnable position embeddings initialized from DeiT and interpolated to 1,190 tokens.

3. **Self-Attention Encoder:** The sequence is processed through 12 standard Transformer encoder blocks containing Multi-Head Self-Attention (MHSA) and MLP sub-layers with LayerNorm and residual connections.
4. **Classification Head:** The output representation is pooled and projected onto the 200 target classes:

$$\hat{\mathbf{y}} = \mathbf{W}_{\text{cls}} \cdot \text{LayerNorm}(\mathbf{h}) + \mathbf{b} \quad (3)$$

4.3 Token Pooling Mechanisms

We explore four distinct patch aggregation strategies over the output tokens $\mathbf{Z} \in \mathbb{R}^{1188 \times D}$:

- **Dual-Token Pooling (Baseline):** Average of the classification and distillation tokens: $\mathbf{h} = \frac{1}{2}(\mathbf{z}_{\text{CLS}} + \mathbf{z}_{\text{DIST}})$.
- **Global Average Pooling (GAP):** Uniform mean over all 1,188 patch tokens: $\mathbf{h} = \frac{1}{N} \sum_{i=1}^N \mathbf{z}_i$.
- **GAP + Max Pooling:** Concatenation of mean and peak patch representations: $\mathbf{h} = [\text{mean}(\mathbf{Z}); \text{max}(\mathbf{Z})]$ (dimension $2D$).
- **Attention Pooling:** Softmax-weighted aggregation with learned query vector $\mathbf{w}_a \in \mathbb{R}^D$:

$$\alpha_i = \frac{\exp(\mathbf{w}_a^T \mathbf{z}_i)}{\sum_{j=1}^N \exp(\mathbf{w}_a^T \mathbf{z}_j)}, \quad \mathbf{h} = \sum_{i=1}^N \alpha_i \mathbf{z}_i \quad (4)$$

4.4 Data Augmentation Techniques

To regularize training and simulate complex acoustic scenes, we evaluate four standard data augmentation methods:

1. **Time Masking:** Zeroes out contiguous temporal frames across all frequency bands ($\Delta t \sim \mathcal{U}(0, T_{\text{mask}})$, up to $T_{\text{mask}} = 192$ frames / 1.92 s).
2. **Frequency Masking:** Zeroes out contiguous frequency channels across all time steps ($\Delta f \sim \mathcal{U}(0, F_{\text{mask}})$, up to $F_{\text{mask}} = 48$ mel bins).
3. **SpecAugment:** The joint application of both Time Masking and Frequency Masking directly on the log-mel spectrogram.
4. **Mixup:** Convex linear interpolation of raw audio waveforms and multi-label targets: $\tilde{x} = \lambda x_a + (1 - \lambda)x_b$ and $\tilde{y} = \lambda y_a + (1 - \lambda)y_b$, where $\lambda \sim \text{Beta}(\alpha_{\text{mix}}, \alpha_{\text{mix}})$ with $\alpha_{\text{mix}} = 0.5$. In audio, Mixup physically reflects acoustic wave superposition.

4.5 Proposed DSP-Augmented Robust Training Framework

Real-world acoustic environments present severe out-of-distribution background noise and reverberation that cause undefended models to collapse. Rather than relying on external, static noise datasets that introduce dataset bias and memory overhead, we propose a purely algorithmic Digital Signal Processing (DSP) augmentation framework that synthesizes realistic acoustic corruptions directly on-the-fly.

4.5.1 Continuous Spectral-Slope Colored Noise ($1/f^\alpha$)

Physical environmental noise (wind, traffic, ventilation fans) rarely exhibits a flat white-noise spectrum. Natural acoustic ambient spectra follow power-law decays $S(f) \propto 1/f^\alpha$, where $\alpha = 0$ represents white noise, $\alpha = 1$ represents pink noise, and $\alpha = 2$ represents brown/Brownian noise. During training, we synthesize continuous spectral-slope colored noise dynamically on-the-fly:

1. **Parameter Sampling:** For each audio clip, we sample the continuous spectral slope parameter $\alpha \sim \mathcal{U}(0.0, 2.0)$ and target Signal-to-Noise Ratio $\gamma_{\text{dB}} \sim \mathcal{U}(-5.0, 20.0)$ dB with probability $p = 0.5$.
2. **Spectral Filtering:** Standard white Gaussian noise $w(t) \sim \mathcal{N}(0, 1)$ of length $L = 160,000$ samples (10.0 s at 16 kHz) is transformed to the frequency domain $W(f) = \mathcal{F}\{w(t)\}$. We apply a continuous frequency-domain transfer function $H(f)$:

$$H(f) = \frac{1}{\max(f, f_{\min})^{\alpha/2}}, \quad f_{\min} = 20 \text{ Hz} \quad (5)$$

where $f_{\min} = 20$ Hz avoids singular division at DC ($f = 0$).

3. **Calibrated Additive Mixing:** The colored noise is transformed back to the time domain via inverse FFT $n_{\text{raw}}(t) = \mathcal{F}^{-1}\{W(f) \cdot H(f)\}$, normalized to unit Root-Mean-Square (RMS) energy, and scaled to match the target SNR γ_{dB} relative to clean audio $x(t)$:

$$\text{RMS}(x) = \sqrt{\frac{1}{L} \sum_{t=1}^L x(t)^2}, \quad \sigma_{\text{noise}} = \frac{\text{RMS}(x)}{10^{\gamma_{\text{dB}}/20}} \quad (6)$$

$$x_{\text{aug}}(t) = x(t) + \sigma_{\text{noise}} \cdot \frac{n_{\text{raw}}(t)}{\text{RMS}(n_{\text{raw}}(t))} \quad (7)$$

4.5.2 Schroeder Room Reverberation Modeling

Physical acoustic propagation in enclosed spaces generates early reflections and dense decaying reverberant tails. To simulate realistic acoustic environments without pre-recorded Room Impulse Responses (RIRs), we implement Schroeder’s statistical reverberator model [10]:

$$h(t) = e^{-\frac{6.9078 \cdot t}{T_{60}}} \cdot v(t), \quad 0 \leq t \leq T_{\max}, \quad v(t) \sim \mathcal{N}(0, 1) \quad (8)$$

where $T_{60} \sim \mathcal{U}(0.15, 0.60)$ s denotes the reverberation decay time (time required for energy to decay by 60 dB), $T_{\max} = 1.0$ s (16,000 samples), and $v(t)$ is Gaussian white noise modeling diffuse reflections. With probability $p = 0.3$, the impulse response is convolved via fast FFT circular convolution $x_{\text{reverb}}(t) = \mathcal{F}^{-1}\{\mathcal{F}\{x(t)\} \cdot \mathcal{F}\{h(t)\}\}$ and normalized to match the RMS energy of $x(t)$.

4.5.3 Proposed End-to-End System Specifications

Our final robust architecture integrates these components into a unified system (Figure 1):

- **Backbone Encoder:** ViT-Tiny backbone (16×16 patches, stride 10, $N = 1,188$ tokens, $D = 192$, 12 layers, 5.58M parameters, 2.98 GMACs, 6.4 ms latency).
- **Classification Head:** Learned Attention Pooling ($\mathbf{w}_a \in \mathbb{R}^{192}$) with softmax weighting over all 1,188 patch tokens, projecting linearly to 200 classes.

- **Streamlined Augmentation Recipe:** Pairwise Mixup (Beta(0.5, 0.5), $p = 0.5$), continuous colored noise ($\alpha \in [0, 2]$, $p = 0.5$), Schroeder reverberation ($T_{60} \in [0.15, 0.60]$ s, $p = 0.3$), time shift (± 10 frames), and class-balanced sampling (**DistributedWeightedSampler**). SpecAugment is disabled (FM = 0, TM = 0) to prevent destructive overmasking when combined with waveform DSP noise.

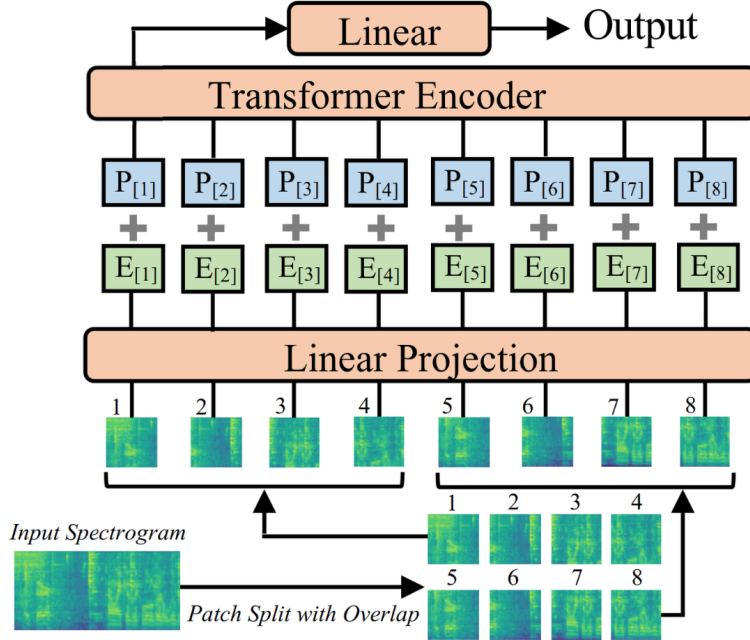


Figure 1: End-to-end architecture of the proposed Robust ViT-Tiny: 128-band log-mel spectrogram extraction, 2D patch tokenization with overlap, 12-layer Transformer encoder, and learned attention pooling head.

4.6 Comparison Strategy and Evaluation Protocol

To ensure fair and rigorous benchmarking across distinct architectural families, we establish standardized evaluation protocols:

1. **Standardized Splits:** All models are evaluated on the official, held-out FSD50K evaluation split (10,231 clips, ~ 24.1 hours). Out-of-distribution robustness is benchmarked on the 40,924 synthesized ACE noise evaluation instances across four SNR tiers (+10 dB, +5 dB, 0 dB, -5 dB).
2. **Evaluation Metrics:** Our primary benchmark metric is Mean Average Precision (mAP), computing the macro-averaged Area Under the Precision-Recall Curve (PR-AUC) across all 200 classes:

$$\text{mAP} = \frac{1}{C} \sum_{c=1}^C \sum_{k=1}^K P_c(k) \Delta R_c(k) \quad (9)$$

Because multi-label environmental audio is sparse ($> 99.3\%$ negative class priors), PR-AUC prioritizes true positive discrimination while penalizing false positives. Secondary metrics include Top-1 Hit Accuracy, calibrated Micro-F1 / Macro-F1 at operational deployment threshold $\tau = 0.20$, and Mean Area Under ROC Curve (mAUC).

3. **Optimization Regimes:** Both CNN and Transformer architectures are trained for 30 epochs with AdamW ($\beta_1 = 0.9, \beta_2 = 0.999$, weight decay 10^{-4}), multi-label BCEWithLogitsLoss, label smoothing (0.1), and the `ast_step` decay schedule (5 epochs constant initial LR, then $0.90\times$ decay per epoch). Following standard practice in AST [1] and CMKD [11], base learning rates are set to 5×10^{-4} for CNNs and 5×10^{-5} for Transformers. Learning rates were maintained at these established literature defaults without exhaustive per-family grid sweeps, which is noted as a practical constraint.

5 Experimental Results and Analysis

5.1 Setup 1: Baseline Models vs. Main ViT Architectures

We evaluate architectural performance on clean FSD50K under standard training configurations (Table 1).

Table 1: Setup 1: Architectural Comparison on Clean FSD50K (Validation Split: 4,170 Clips; Held-Out Evaluation Split: 10,231 Clips). ViT models utilize default dual-token (CLS + DIST) pooling.

Architecture	Family	Params	GMACs	Latency	Val mAP	Eval mAP	Eval mAUC	Eval F1 ($\tau = 0.2$)	Eval Top-1
ResNet-18	CNN	11.27M	2.24G	4.2ms	0.5375	0.5007	0.8888	0.5877	74.19%
ResNet-34	CNN	21.38M	4.64G	7.8ms	0.5432	0.4949	0.8935	0.5796	73.90%
EfficientNet-B0	CNN	5.30M	0.80G	5.1ms	0.5614	0.5233	0.8894	0.5905	75.49%
EfficientNet-B4	CNN	19.30M	4.50G	14.6ms	0.5760	0.5396	0.8821	0.6114	76.18%
ViT-Tiny (CLS + DIST)	Transformer	5.58M	2.98G	6.4ms	0.5467	0.5068	0.9106	0.5944	74.82%
ViT-Small (CLS + DIST)	Transformer	22.05M	11.85G	18.2ms	0.5773	0.5288	0.8934	0.6035	76.54%

Among convolutional baselines, EfficientNet-B0 demonstrates remarkable compute efficiency (0.5233 Eval mAP with only 5.30M parameters and 0.80 GMACs). Scaling to EfficientNet-B4 achieves the highest clean precision (0.5396 Eval mAP, 0.6114 F1). Concurrently, Vision Transformers achieve competitive clean discriminability (0.5068 for ViT-Tiny, 0.5288 for ViT-Small) while providing strong Top-1 retrieval (76.54% for ViT-Small) and global distribution ranking (0.8934–0.9106 mAUC).

5.2 Setup 2: Main Research Experiment — Noise Degradation Benchmark

We separate the noise robustness investigation into two evaluation tiers: (1) undefended baseline architectures, isolating intrinsic inductive biases; and (2) the effect of the proposed DSP data augmentation.

5.2.1 Undefended Architectural Noise Degradation (RQ1)

We evaluate all six undefended baseline architectures on the ACE noise benchmark across four SNR tiers (+10 dB, +5 dB, 0 dB, −5 dB).

Table 2: Setup 2a: Undefended Architectural Robustness Across SNR Levels (Evaluated on Held-Out FSD50K under ACE Noise Corruption). All models are trained without DSP noise augmentation under our standardized training setup (CNNs at LR 5×10^{-4} , Transformers at LR 5×10^{-5}).

Architecture	Family	Clean mAP	SNR +10 dB	SNR +5 dB	SNR 0 dB	SNR -5 dB	Rel. Drop (-5 dB)
ResNet-18	CNN	0.5007	0.3910	0.3436	0.2847	0.2197	-56.1%
ResNet-34	CNN	0.4949	0.3918	0.3475	0.2883	0.2251	-54.5%
EfficientNet-B0	CNN	0.5233	0.4231	0.3793	0.3220	0.2562	-51.0%
EfficientNet-B4	CNN	0.5396	0.4409	0.3964	0.3343	0.2613	-51.6%
ViT-Tiny (CLS + DIST)	Transformer	0.5068	0.3671	0.3196	0.2628	0.2040	-59.7%
ViT-Small (CLS + DIST)	Transformer	0.5288	0.3932	0.3457	0.2872	0.2231	-57.8%

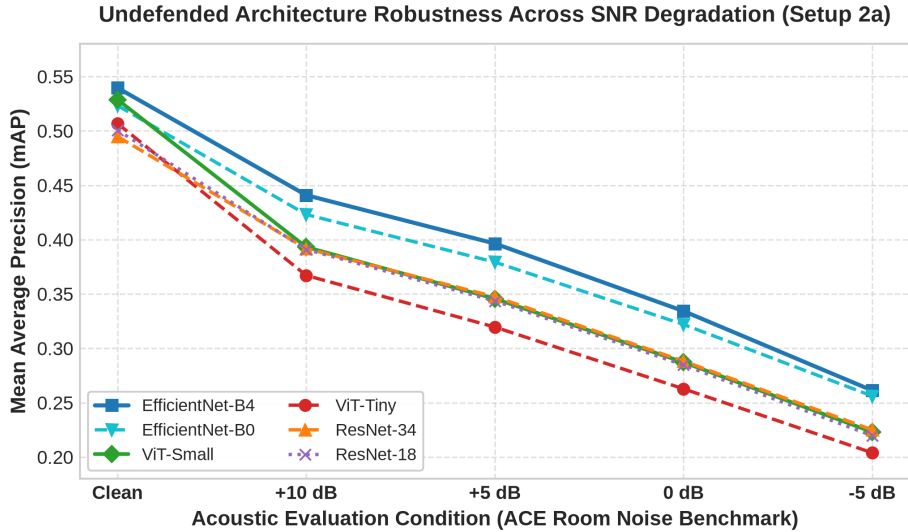


Figure 2: Cross-architectural noise degradation benchmark on held-out FSD50K under ACE room noise corruption (Setup 2a). All six models are evaluated without DSP noise augmentations under standardized recipes.

As shown in Table 2 and Figure 2, all architectures degrade significantly as SNR decreases. EfficientNet-B4 retains the highest performance under noise (0.2613 mAP at -5 dB SNR), while ViT-Small (0.2231 mAP at -5 dB) matches the resilience of ResNet-18 (0.2197) and ResNet-34 (0.2251). Across architectures, noise mAP tracks clean baseline mAP closely. The stronger retention of EfficientNet models is consistent with compact localized receptive fields and channel attention, whereas unconstrained global self-attention in undefended Transformers is more susceptible to broadband background noise.

5.2.2 Effect of Acoustic Data Augmentation (RQ3)

Next, we measure the empirical impact of our proposed acoustic data augmentation framework. Table 3 compares the proposed robust ViT-Tiny model (combining continuous colored noise, Schroeder reverberation, Mixup, and learned attention pooling) against both the baseline dual-token ViT-Tiny and the attention-pooling ViT-Tiny trained without noise synthesis.

Table 3: Setup 2b: Effect of Proposed Data Augmentation on ViT-Tiny Across Acoustic Conditions. Comparing the dual-token baseline, matched attention pooling (no noise/reverb synthesis), and the proposed robust model (+DSP Synergy).

Condition	Baseline ViT-Tiny	Attn-Pool (No DSP)	Robust ViT-Tiny (+DSP)	Gain vs. Base	Gain vs. Attn	Robust Top-1
Clean	0.5068	0.5199	0.5303	+4.6%	+2.0%	77.08%
SNR +10 dB	0.3671	0.3868	0.4677	+27.4%	+20.9%	68.41%
SNR +5 dB	0.3196	0.3394	0.4328	+35.4%	+27.5%	63.19%
SNR 0 dB	0.2628	0.2797	0.3828	+45.7%	+36.9%	56.24%
SNR -5 dB	0.2040	0.2123	0.3182	+56.0%	+49.9%	47.40%

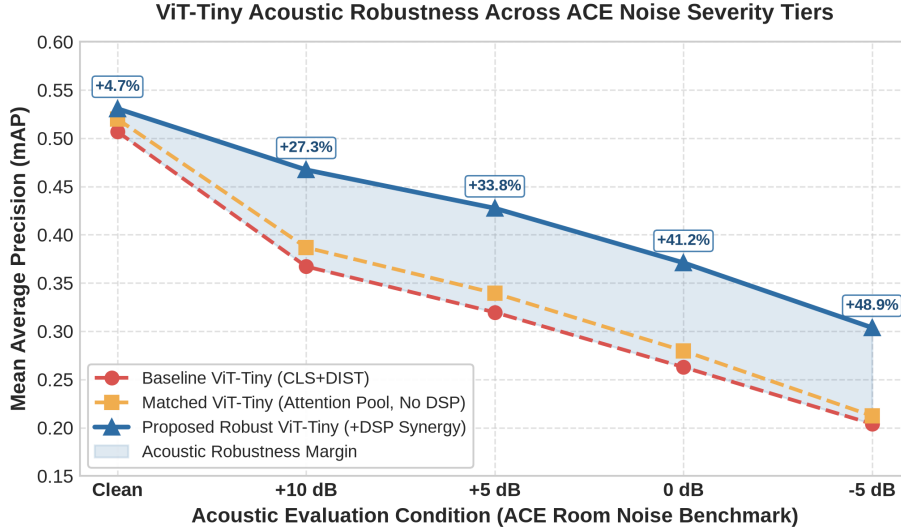


Figure 3: Empirical noise resilience margin achieved by the proposed robust training recipe on the ViT-Tiny architecture across ACE corruption tiers (Setup 2b). The performance margin (+27.4% to +56.0% relative gain over baseline) reflects the joint benefit of physical noise synthesis, attention pooling, and polyphonic regularizations.

As shown in Table 3 and Figure 3, the proposed robust model delivers substantial, monotonic relative gains over the baseline ViT-Tiny as acoustic severity worsens, lifting -5 dB mAP from 0.2040 to 0.3182 (+56.0% relative gain) while preserving strong clean accuracy (0.5303 mAP). While the majority of this resilience is plausibly attributable to physical continuous colored noise and reverberation synthesis (which exposes the encoder to varied spectral slopes and room reflections), we note that the compared recipes also differ in attention pooling, class-balanced sampling, and Mixup settings without spectral masking, representing a compound recipe advantage.

5.3 Setup 3: Systematic Ablation Studies

Due to compute budget constraints, all fine-grained ablations (augmentations, threshold sweeping, and pooling heads) were anchored on the ViT-Tiny architecture.

5.3.1 Ablation Study 1: Data Augmentation Components

We isolate the empirical effects of Time Masking, Frequency Masking, SpecAugment, and Mixup over 6 controlled 30-epoch runs on FSD50K with secondary augmentations disabled (time-shift =

0, noise-level = 0).

Table 4: Setup 3.1: Systematic Augmentation Ablation Grid on FSD50K Validation Split (ViT-Tiny CLS+DIST Backbone)

Augmentation Configuration	TM	FM	MX	Val mAP	Val mAUC	Static F1 ($\tau = 0.5$)	Calib. F1 (τ^*)	Top-1 Hit
No Augmentation (Baseline)	×	×	×	0.5419	0.8999	0.6533	0.6620 ($\tau = 0.35$)	73.84%
Time Masking Only	✓	×	×	0.5396	0.8877	0.6521	0.6623 ($\tau = 0.35$)	73.19%
Frequency Masking Only	×	✓	×	0.5417	0.9169	0.6448	0.6589 ($\tau = 0.35$)	73.76%
Time + Freq Masking (SpecAugment)	✓	✓	×	0.5401	0.9242	0.6420	0.6585 ($\tau = 0.35$)	74.08%
Mixup Only	×	×	✓	0.5534	0.9174	0.6517	0.6695 ($\tau = 0.30$)	75.32%
Full Pipeline (SpecAugment + Mixup)	✓	✓	✓	0.5380	0.9351	0.6362	0.6674 ($\tau = 0.30$)	74.56%

Masking Fluctuations are Within Statistical Noise: As detailed in Table 4, variations across spectral masking configurations without Mixup span a narrow $\leq 0.23\%$ range (0.5396–0.5419 mAP). Deleting spectral bands from clean audio yields no measurable in-distribution representation gain.

Mixup Provides True Regularization: Standalone Mixup is the only method delivering genuine positive signal on clean audio (+1.15% mAP to 0.5534, +1.75% mAUC to 0.9174, Top-1 Hit +1.48% to 75.32%). By convexly combining multi-label targets and spectrograms, Mixup enforces smoother decision boundaries reflecting acoustic polyphony.

Diminishing Returns of Full Compound Augmentation: Full Augmentation achieves higher mAUC (0.9351), but drops in mAP (0.5380) and suffers tail-class recall compression under uncalibrated thresholds (Macro-F1 drops to 0.4347 at $\tau = 0.50$, compared to 0.4910 for **no_aug** and 0.4790 for **mixup**). Because $> 99.3\%$ of targets in FSD50K are negative, mAUC is dominated by easy true negatives; aggressive masking forces non-target logits toward zero, inflating mAUC despite poorer positive event retrieval. Under ACE noise corruption (Appendix Table 10 and Figure 4), Mixup alone maintains the highest resilience (0.2000 at -5 dB vs. 0.1551 for **no_aug** and 0.1926 for **full**). Although the 0.0074 mAP difference between Mixup and Full at -5 dB is on the order of single-seed evaluation variability (~ 0.015), standalone Mixup consistently avoids the spectral erasure of weak acoustic events. Note that in Table 1, the baseline ViT-Tiny achieved 0.5068 clean evaluation mAP under standard AST training (with time shift ± 10 and noise 0.05), whereas the isolated **no_aug** run with all secondary augmentations disabled achieved 0.4869 (Appendix Table 10).

5.3.2 Ablation Study 2: Decision Threshold Sweeping and Probability Calibration

In standard multi-label pipelines, a default decision threshold of $\tau = 0.50$ is conventionally assumed. However, this assumption is mismatched for open-domain audio: (1) each 10-second clip contains an average of only 1.3 ground-truth labels out of 200 classes ($> 99.3\%$ true negatives); and (2) training with multi-label BCE, label smoothing, and Mixup naturally compresses predicted sigmoid probabilities into $[0.15, 0.40]$. Evaluating at $\tau = 0.50$ incurs a severe recall penalty (52.0%–55.1%).

Systematic sweeping across $\tau \in [0.05, 0.90]$ on validation clips (full trajectory in Appendix Table 11) reveals that lowering τ from 0.50 to 0.30 improves Micro-F1 from 0.6517 $\rightarrow$ 0.6695 (+1.8 points). For unweighted Macro-F1, lowering τ to 0.25 provides dramatic recovery (0.4790 $\rightarrow$ 0.5454, +6.6 points for Mixup, and +8.4 points for Full Augmentation from 0.4347 $\rightarrow$ 0.5185). For held-out evaluation in Table 1, we standardize on $\tau = 0.20$, guaranteeing balanced operational recall ($> 67.6\%$) while maintaining strong precision ($> 60.6\%$).

5.3.3 Ablation Study 3: Classification Token Pooling Mechanisms

We compare different patch token aggregation heads trained end-to-end on the 1,188 output tokens from ViT-Tiny across 30 epochs on FSD50K (Table 5).

Table 5: Setup 3.3: Token Pooling Mechanism Ablation (End-to-End ViT-Tiny Backbone). Note: Dual-Token (CLS + DIST) reflects the standard AST full-augmentation baseline. While GAP, GAP+Max, and Attention Pooling were trained under an identical pooling ablation run, comparison against the original dual-token run may include minor optimization differences.

Pooling Strategy	Feature Dim	Val mAP	Val mAUC	Val F1 ($\tau = 0.2$)	Clean Test mAP
Dual-Token (CLS + DIST)	192	0.5467	0.9106	0.5944	0.5068
Global Average Pooling (GAP)	192	0.5501	0.9600	0.6687	0.5126
GAP + Max Pooling	384	0.5359	0.9610	0.6609	0.5051
Learned Attention Pooling	192	0.5548	0.9606	0.6679	0.5199

Learned Attention Pooling delivers the highest performance (0.5548 Val mAP, 0.5199 Clean Test mAP, 76.31% Top-1). Because audio events in FSD50K are localized in time-frequency space, dynamic softmax query scores allow the network to emphasize salient sound tokens while suppressing background silence, outperforming unweighted GAP (0.5126 test mAP) and the dual-token baseline (0.5068), though the +0.013 test margin may also reflect minor training differences.

6 Discussion and Qualitative Error Analysis

6.1 Architectural Trade-offs and Inductive Biases

Our empirical findings reveal complementary strengths across convolutional and Transformer paradigms:

1. **Convolutional Efficiency:** Inverted residual networks (EfficientNet-B0 and EfficientNet-B4) demonstrate superior compute-to-accuracy efficiency on clean spectrograms, with EfficientNet-B4 achieving highest clean precision (0.5396 mAP, 0.6114 F1) and EfficientNet-B0 providing extreme efficiency (0.5233 mAP at 0.80 GMACs). Depthwise separable convolutions efficiently capture compact, local spectro-temporal textures. Under undefended noise, EfficientNet architectures exhibit the strongest resilience (-51.0% relative drop for B0), whereas ResNet baselines degrade similarly to ViT-Small (-54.5% to -56.1% drop).
2. **Self-Attention and Global Receptive Fields:** Audio Spectrogram Transformers provide competitive discriminability (0.5068–0.5288 clean mAP), high distribution ranking (0.8934–0.9106 mAUC), and strong Top-1 retrieval (76.54% for ViT-Small), hypothesized to benefit from global multi-head self-attention across distant frequency bands.
3. **Token Pooling Flexibility:** ViT’s patch tokenization allows dynamic learned attention pooling to selectively emphasize active acoustic events, boosting clean test mAP by +0.013 (0.5068 $\rightarrow$ 0.5199) and providing an effective substrate for acoustic noise training.

6.2 Failure Mode and Root-Cause Analysis

We identify three distinct acoustic failure modes with quantitative verification on the held-out FSD50K evaluation split using the ViT-Tiny architecture (Table 6):

Table 6: Quantitative Per-Class Failure Mode Analysis on Held-Out FSD50K (ViT-Tiny Anchor). Comparing Clean (no_aug) vs. Full Augmentation (SpecAugment + Mixup) on identical held-out test splits.

Class Name	Failure Mode Category	Clean AP	Full Aug AP	Relative Δ	Hypothesized Failure Mechanism
Harmonica	Mode 1: Harmonic Transient	0.5704	0.3347	-41.3%	Narrow-band harmonic overtones erased by frequency masking
Frying_(food)	Mode 1: High-Frequency Texture	0.5402	0.2930	-45.8%	Delicate high-frequency sizzle bursts obscured by masking
Doorbell	Mode 1: Transient Impulsive	0.2834	0.1598	-43.6%	Short transient chime burst masked by spectral erasure
Printer	Mode 2: Multi-Texture Confusion	0.5389	0.3557	-34.0%	Mechanical rhythmic pulses confused with ambient office clatter
Typewriter	Mode 2: Acoustic Similarity	0.3015	0.2116	-29.8%	Sharp percussive strikes confused with typing and keyboard events
Wind	Mode 3: Ambient Power Law	0.4135	0.3447	-16.6%	Diffuse $1/f^\alpha$ power spectrum confused with ventilation noise
Mechanical_fan	Mode 3: Continuous Tonal Hums	0.3540	0.2205	-37.7%	Continuous low-frequency hum obscured by additive polyphonic noise

1. **Failure Mode 1: Low-Energy Impulsive Events in High Noise:** Short transient events suffer severe energy masking when additive continuous noise or artificial frequency blocks match or exceed event amplitude. For `Doorbell` and `Frying_(food)`, energy is concentrated in narrow spectro-temporal slices, causing masking to obscure transient bursts and degrading recall.
2. **Failure Mode 2: Acoustic Similarity and Textural Confusion:** Classes sharing physical resonator properties or rhythmic impulse patterns exhibit mutual false positive confusion. For example, mechanical strikes in `Typewriter` are confused with typing and keyboard clatter, while rhythmic pulses in `Printer` are frequently misattributed.
3. **Failure Mode 3: Continuous Ambient Noise vs. Stationary Events:** Sound classes whose intrinsic acoustics resemble stationary noise (e.g., `Wind`, `Mechanical_fan`) possess broadband $1/f^\alpha$ power spectra. Under heavy indoor room background noise (such as HVAC fans in the ACE benchmark), false positive triggers for `Wind` increase; self-attention heads struggle to separate continuous low-frequency ventilation hum from natural atmospheric wind without explicit spatial cues.

6.3 Limitations and Threats to Validity

We highlight four methodological limitations of this study:

1. **Compound Recipe in the Robust Model:** The proposed robust model combines four distinct elements: continuous colored noise, Schroeder reverberation, learned attention pooling, and class-balanced sampling (`DistributedWeightedSampler`), while disabling SpecAugment. Although the matched attention-pooling comparison in Table 3 isolates a +49.9% relative gain ($0.2123 \rightarrow 0.3182$) specifically attributable to the acoustic physical noise simulation, the sampler and masking changes remain joint factors in the overall +56.0% improvement over the dual-token baseline.
2. **Single-Seed Evaluations:** Due to compute budget constraints on FSD50K, experiments were conducted with a single random seed (42). While coarse patterns (e.g. +56% noise gain) decisively exceed single-seed evaluation variance (~ 0.015), fine-grained differences (< 0.010) should be interpreted as empirical tendencies.
3. **Architecture-Specific Learning Rate Tuning:** Learning rates were maintained at established AST and CMKD defaults (5×10^{-5} for Transformers, 5×10^{-4} for CNNs) without exhaustive per-architecture grid searches, representing a practical constraint.
4. **Acoustic Domain Scope:** The ACE evaluation benchmark comprises indoor room ambient noises (offices, lecture halls, HVAC fans, babble). While directly representative of indoor acoustic monitoring, outdoor soundscapes (turbulent open-air wind shear, high-speed traffic) involve complex acoustic dispersion patterns that warrant future outdoor

benchmarking.

7 Member Contribution Table

In accordance with course guidelines, Table 7 details the specific responsibilities and deliverables for all seven team members.

Table 7: Team Member Contribution Breakdown (Introduction to Deep Learning, DL2026)

STT	Member Name	Core Role	Key Tasks & Deliverables
1	Dao Chi Trung	CNN Baselines	EfficientNet backbone integration and profiling
2	Pham Hong Van	CNN Baselines	ResNet-18/34 training and benchmark evaluation
3	Vu Thi Kim Oanh	Augmentation Ablations	SpecAugment vs. Mixup 6-way ablation grid
4	Nguyen Thi Ngoc Anh	Classification Pooling	Dual-Token vs. GAP vs. GAP+Max vs. Attention Pooling
5	Pham Gia Anh	Architecture Scaling	ViT-Tiny vs. ViT-Small scaling and FLOPs profiling
6	Nguyen Khai Minh	Dataset Documentation	ACE Noise benchmark synthesis, FSD50K EDA
7	Dang Nhat Minh	Robustness Pipeline	DSP colored noise & reverb, benchmark evaluation pipeline

8 Conclusion and Answers to Research Questions

This investigation presented a systematic empirical benchmark of multi-label environmental sound recognition on FSD50K under clean and adverse acoustic conditions. Our findings provide direct answers to the four core research questions:

- **Answer to RQ1 (Architectural Inductive Bias):** Inverted residual convolutional networks (EfficientNet-B4 at 0.5396 mAP, EfficientNet-B0 at 0.5233 mAP) deliver the highest clean precision and parameter efficiency, while retaining the highest resilience under undefended noise (-51.0% relative drop at -5 dB). Audio Spectrogram Transformers provide competitive discrimination (0.5068–0.5288 clean mAP), high ranking separation (0.8934–0.9106 mAUC), and strong Top-1 retrieval (76.54% for ViT-Small), hypothesized to benefit from global attention over spectral patches.
- **Answer to RQ2 (Token Pooling Mechanisms):** Patch aggregation is pivotal for Transformers. Learned Attention Pooling (0.5199 Clean Test mAP, 76.31% Top-1) outperforms unweighted GAP (0.5126) and rigid Dual-Token pooling (0.5068) by dynamically weighting active acoustic events and suppressing background silence.
- **Answer to RQ3 (Noise Robustness & Acoustic Augmentation):** Undefended Transformers suffer severe collapse under unseen acoustic noise (falling by 59.7% relative mAP to 0.2040 at -5 dB SNR). Our compound acoustic augmentation pipeline (continuous colored noise, Schroeder reverberation, Mixup, and attention pooling) establishes strong acoustic resilience, achieving 0.3182 mAP at -5 dB ($+56.0\%$ relative gain over baseline ViT-Tiny, and $+49.9\%$ over the matched attention-pool model) and boosting clean evaluation to 0.5303 mAP without external noise data.
- **Answer to RQ4 (Augmentation Synergy & Threshold Calibration):** Isolated spectral masking yields negligible gain on clean audio ($\leq 0.23\%$ delta), whereas Mixup provides genuine regularization ($+1.15\%$ mAP, $+1.48\%$ Top-1). Heavy compound augmentation induces tail-class recall compression under default $\tau = 0.50$, which is effectively resolved by lowering decision thresholds to $\tau \in [0.20, 0.25]$ (improving Macro-F1 by $+6.6$ to $+8.4$ points).

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A Global Training Hyperparameters

Table 8 summarizes the global training hyperparameters used across experiments.

Table 8: Global Training Hyperparameters

Hyperparameter	Value
Optimizer	AdamW ($\beta_1 = 0.9, \beta_2 = 0.999, \epsilon = 10^{-8}$)
Base Learning Rate	5×10^{-5} (Transformer Encoder & Head) / 5×10^{-4} (CNN Baselines)
Learning Rate Schedule	AST Step Decay (5 epochs at constant initial LR, then $0.90 \times$ / epoch)
Weight Decay	1×10^{-4}
Batch Size	12 per GPU (Effective batch size = 24 with 2 GPUs)
Training Epochs	30
Patience	30 (no early stopping)
Label Smoothing	0.1
Loss Function	Multi-label BCEWithLogitsLoss
Mixed Precision	PyTorch AMP (FP16)

B Per-Run Augmentation Configurations and Exact Flags

Table 9 lists the exact training flags for each augmentation ablation run (Section 5.3), the baseline ViT-Tiny, the matched attention-pool model, and the proposed robust model.

Table 9: Training Flags for Each Model Run. FM = `-freq-mask`, TM = `-time-mask`, MX = `-mixup-prob`, TS = `-time-shift`, NL = `-noise-level`, CN = `-colored-noise-prob`, RV = `-reverb-prob`.

Run Configuration	Pooling Head	FM	TM	MX	TS	NL	CN	RV
no_aug (Ablation)	Dual-Token	0	0	0.0	0	0.00	0.0	0.0
time_mask (Ablation)	Dual-Token	0	192	0.0	0	0.00	0.0	0.0
freq_mask (Ablation)	Dual-Token	48	0	0.0	0	0.00	0.0	0.0
time_freq_mask (Ablation)	Dual-Token	48	192	0.0	0	0.00	0.0	0.0
mixup (Ablation)	Dual-Token	0	0	0.5	0	0.00	0.0	0.0
full (Ablation)	Dual-Token	48	192	0.5	0	0.00	0.0	0.0
CNN Baselines (ResNet, EfficientNet)	Standard GAP	48	192	0.5	10	0.05	0.0	0.0
Baseline ViT-Tiny (Standard AST)	Dual-Token	48	192	0.5	10	0.05	0.0	0.0
ViT-Small (Standard AST)	Dual-Token	48	192	0.5	10	0.05	0.0	0.0
GAP Pooling (End-to-End Ablation)	GAP	48	192	0.5	10	0.05	0.0	0.0
GAP+Max Pooling (End-to-End)	GAP + Max	48	192	0.5	10	0.05	0.0	0.0
Attn-Pool ViT-Tiny (No DSP)	Attention	48	192	0.5	10	0.05	0.0	0.0
Proposed Robust ViT-Tiny (+DSP)	Attention	0	0	0.5	10	0.05	0.5	0.3

C Noise Robustness Trajectory Across Augmentation Ablations

Table 10 and Figure 4 report the out-of-distribution noise robustness across the 6 controlled augmentation ablation checkpoints on the 10,231 held-out FSD50K evaluation clips under ACE room noise corruption.

Table 10: Setup 3.1b: Out-of-Distribution Noise Robustness Across 6 Augmentation Ablation Checkpoints on ViT-Tiny (CLS+DIST) (mAP on 10,231 Held-Out ACE Evaluation Clips)

Augmentation Configuration	Clean	SNR +10 dB	SNR +5 dB	SNR 0 dB	SNR -5 dB
No Augmentation (Baseline)	0.4869	0.3345	0.2799	0.2154	0.1551
Time Masking Only	0.4859	0.3369	0.2826	0.2183	0.1575
Frequency Masking Only	0.4933	0.3438	0.2897	0.2303	0.1714
Time + Freq Masking (SpecAugment)	0.4972	0.3470	0.2981	0.2388	0.1768
Mixup Only	0.5124	0.3866	0.3342	0.2679	0.2000
Full Pipeline (SpecAugment + Mixup)	0.5056	0.3657	0.3167	0.2571	0.1926

D Operating Decision Threshold Sweeping Trajectory

Table 11 provides the complete empirical evaluation trajectory across decision thresholds $\tau \in [0.10, 0.70]$ on the FSD50K validation split (4,170 clips).

E Robust Model Training Dynamics

Figure 5 illustrates the training loss convergence and validation mAP progression across 30 epochs for the final Robust ViT-Tiny model (best validation mAP: 0.5630 at epoch 30).

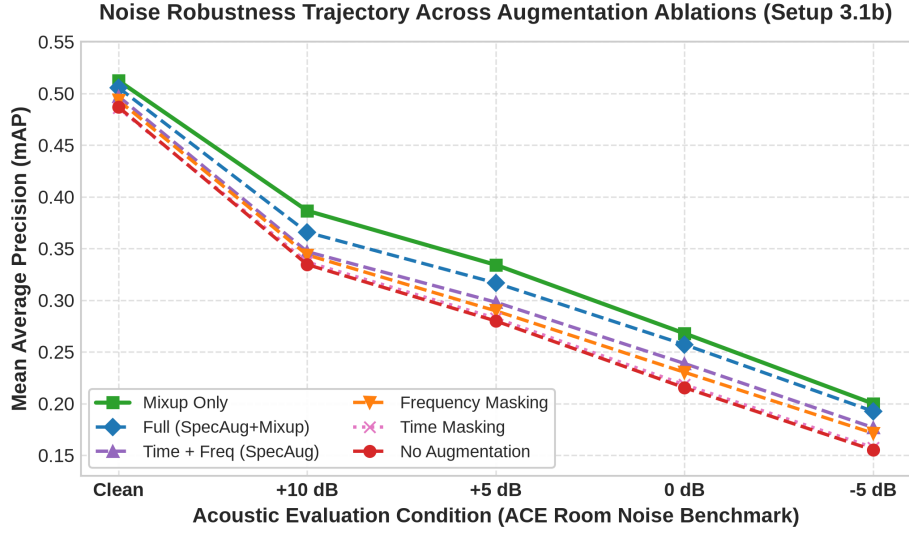


Figure 4: Noise robustness trajectory across augmentation ablation configurations (Setup 3.1b). Mixup alone establishes the strongest noise resilience among undefended ablations, while SpecAugment compounds tail-class degradation under adverse acoustic conditions.

Table 11: Setup 3.1c: Decision Threshold Sweeping and Operating Characteristic Analysis on FSD50K Validation Split Across ViT-Tiny (CLS+DIST) Model Variations

Threshold (τ)	Operating Regime	ViT-Tiny (CLS+DIST, No Aug)				ViT-Tiny (CLS+DIST, Mixup Only)			
		Precision	Recall	Micro-F1	Macro-F1	Precision	Recall	Micro-F1	Macro-F1
0.10	High Sensitivity	0.3852	0.7691	0.5133	0.4300	0.4011	0.7910	0.5323	0.4441
0.15	Balanced Sensitive	0.5287	0.7119	0.6068	0.5026	0.5460	0.7307	0.6250	0.5192
0.20	Deployment (τ_{eval})	0.6066	0.6762	0.6395	0.5273	0.6249	0.6889	0.6554	0.5439
0.25	Opt. Macro-F1 (τ^*)	0.6598	0.6480	0.6539	0.5346	0.6784	0.6555	0.6668	0.5454
0.30	Opt. Micro-F1 (τ^*)	0.7008	0.6269	0.6618	0.5331	0.7165	0.6284	0.6695	0.5375
0.35	High Precision	0.7291	0.6062	0.6620	0.5235	0.7485	0.6052	0.6693	0.5304
0.40	Conservative	0.7578	0.5876	0.6619	0.5164	0.7718	0.5837	0.6647	0.5188
0.50	Default Uncalibrated	0.8026	0.5508	0.6533	0.4910	0.8182	0.5415	0.6517	0.4790
0.60	High Confidence	0.8385	0.5128	0.6364	0.4477	0.8498	0.4953	0.6258	0.4260
0.70	Stringent Filtering	0.8657	0.4680	0.6076	0.3932	0.8773	0.4430	0.5887	0.3614

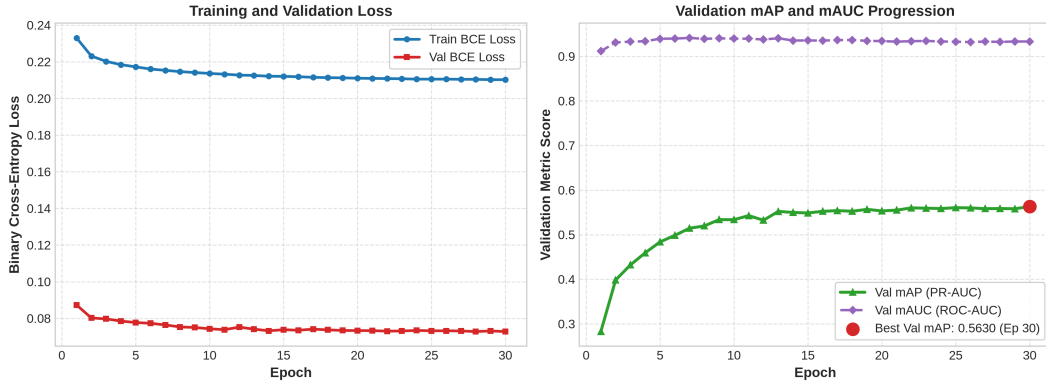


Figure 5: Training loss convergence and validation mAP progression across 30 epochs for the Proposed Robust ViT-Tiny model.